

ED + MCMC Direction Reconstruction

3 Categories, 120 Events

PERFORMANCE SUMMARY

Cosine of Angle (True vs Reconstructed Direction)

Mean: 0.0519
Median (p50): 0.1139
68th percentile: 0.4863 ← PRIMARY METRIC
90th percentile: 0.7995
Std deviation: 0.5734

Angular Error (degrees)

Mean: 86.36°
Median: 83.46°
68th percentile: 109.68°
90th percentile: 134.47°

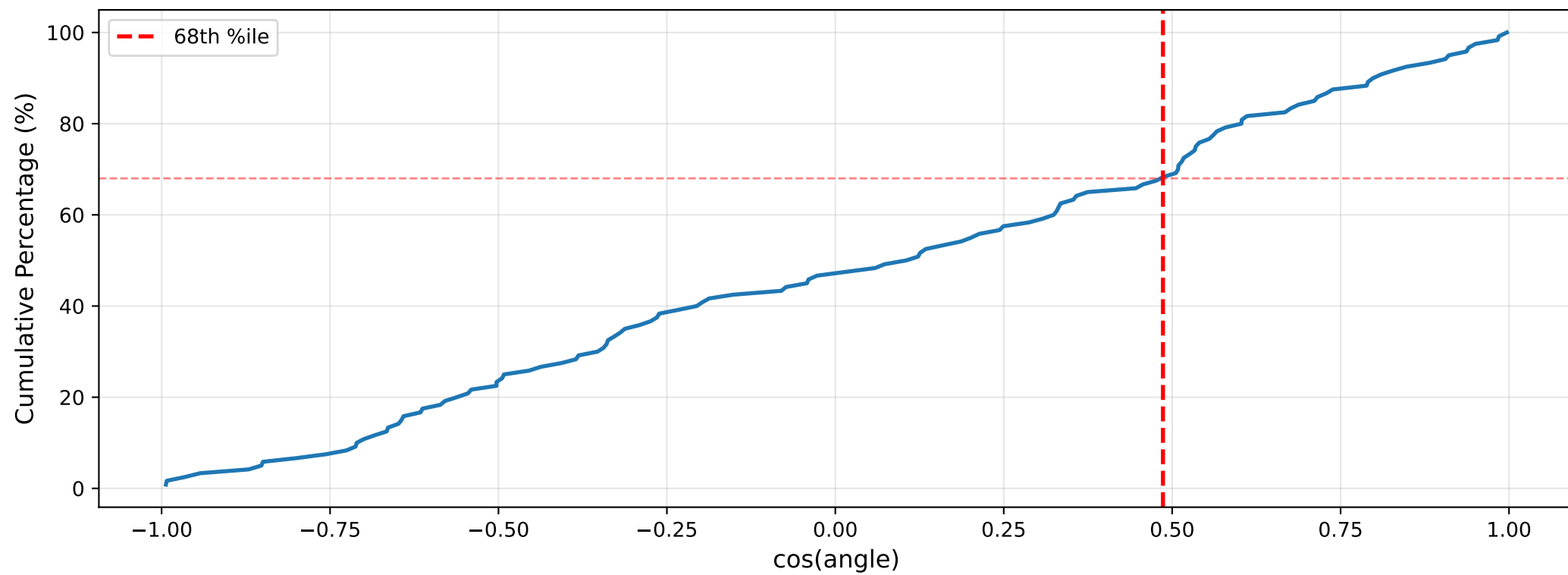
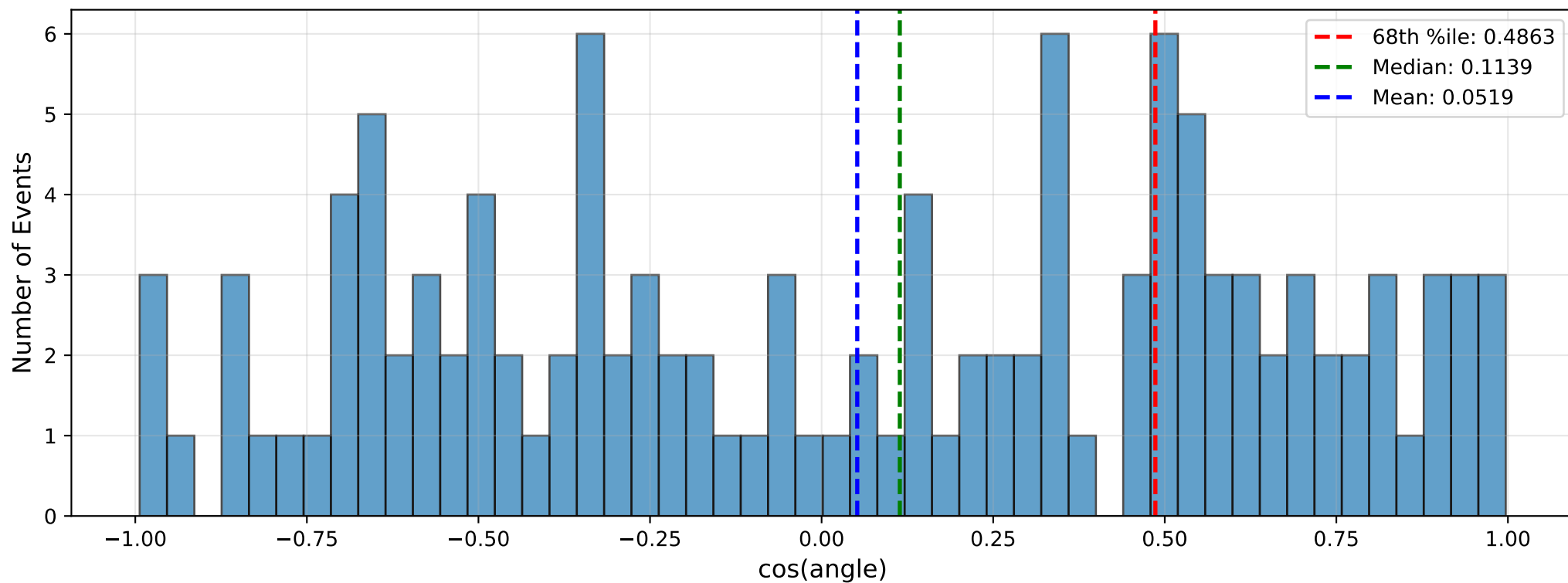
INTERPRETATION

68% of events have $\cos(\text{angle}) \geq 0.4863$
→ 68% of events within 60.90° angular resolution

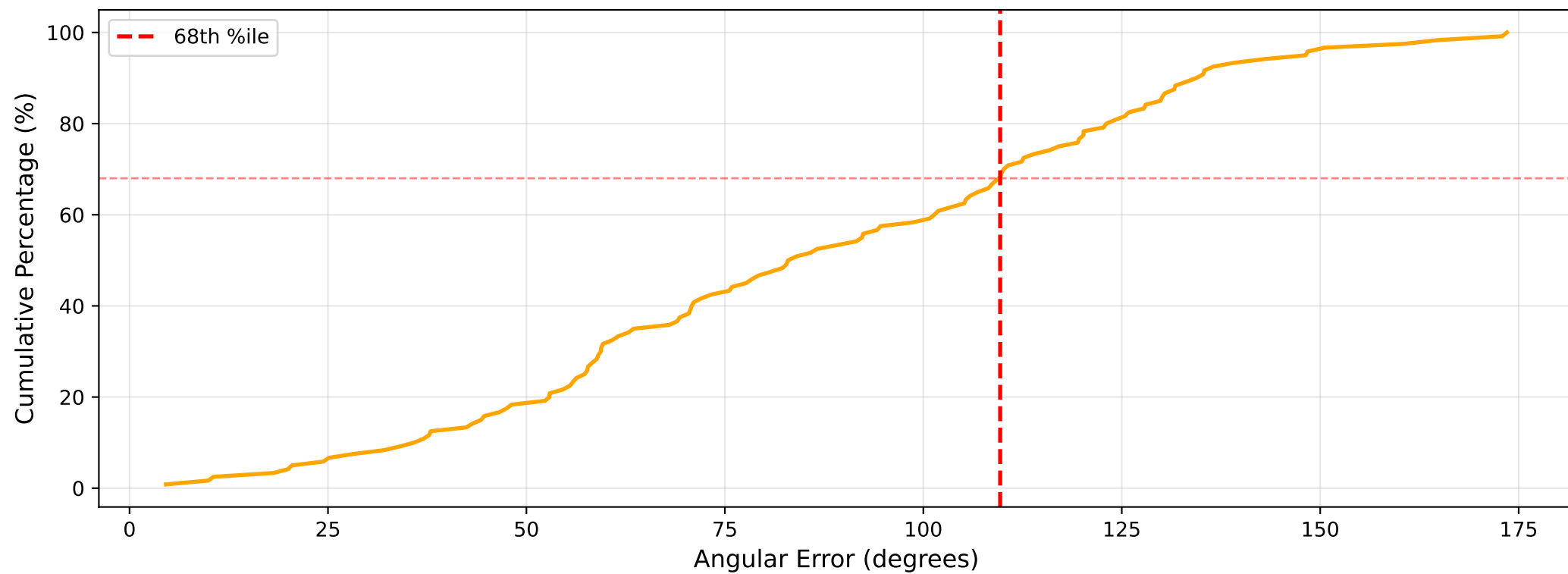
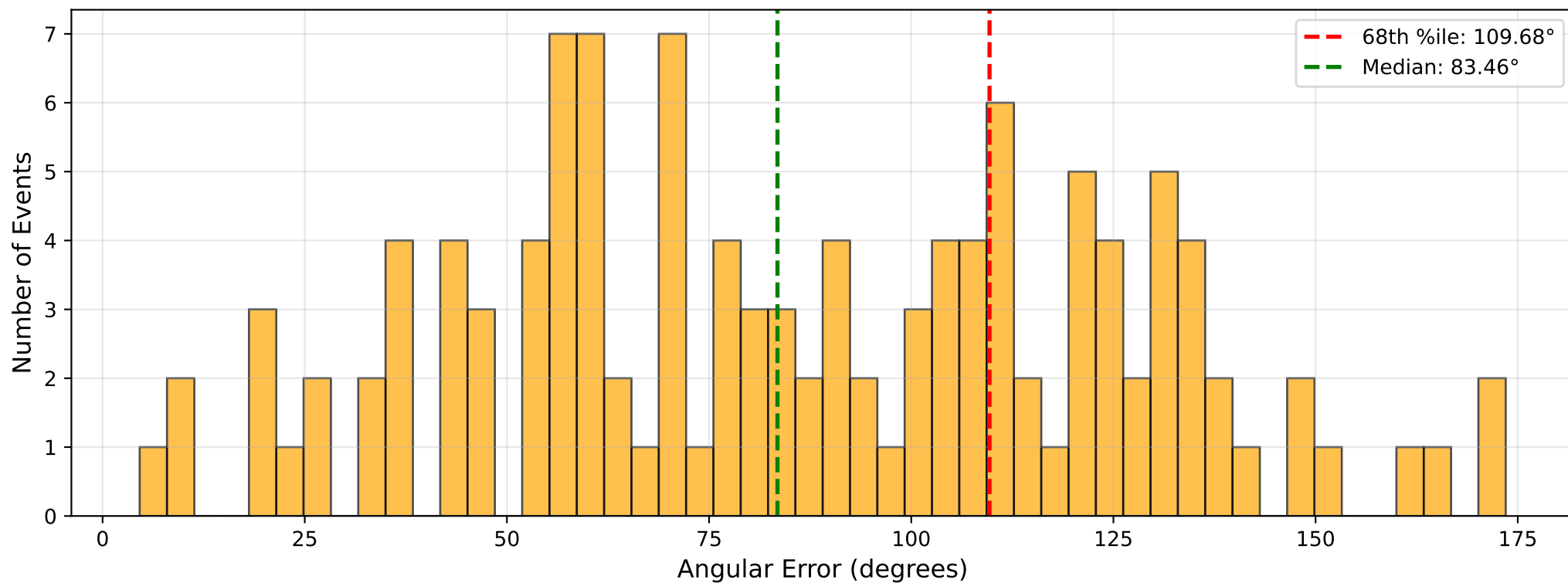
Perfect pointing: $\cos(\text{angle}) = 1.0$ (0°)
Random pointing: $\cos(\text{angle}) \sim 0.0$ (90°)

Dataset: 3 categories × ~40 events/cat

Distribution of $\cos(\text{angle})$ Between True and Reconstructed



Angular Error Distribution (degrees)



Performance by Category

